Rule v8: the index calculus against a matched rho

Two runs of one binary (`ic` from commit 370e1919), six sizes of ledger §23, 64 targets a size, 408 rows each, Emerald Rapids container, isolated one-core runs. **v8**: rho's signed-Frobenius canonicalisation batched 16 lanes at a time. **v8c**: the same binary with `KIC_RHO_CANON=scalar` (rho as at rule v7). Source: `matched.json`, `analysis.json`, `PROTOCOL.md` (declared before any row). Class: matched rho = engineering (rho side); larger distance = accounting; slope = measurement. No advance; no index-calculus improvement.

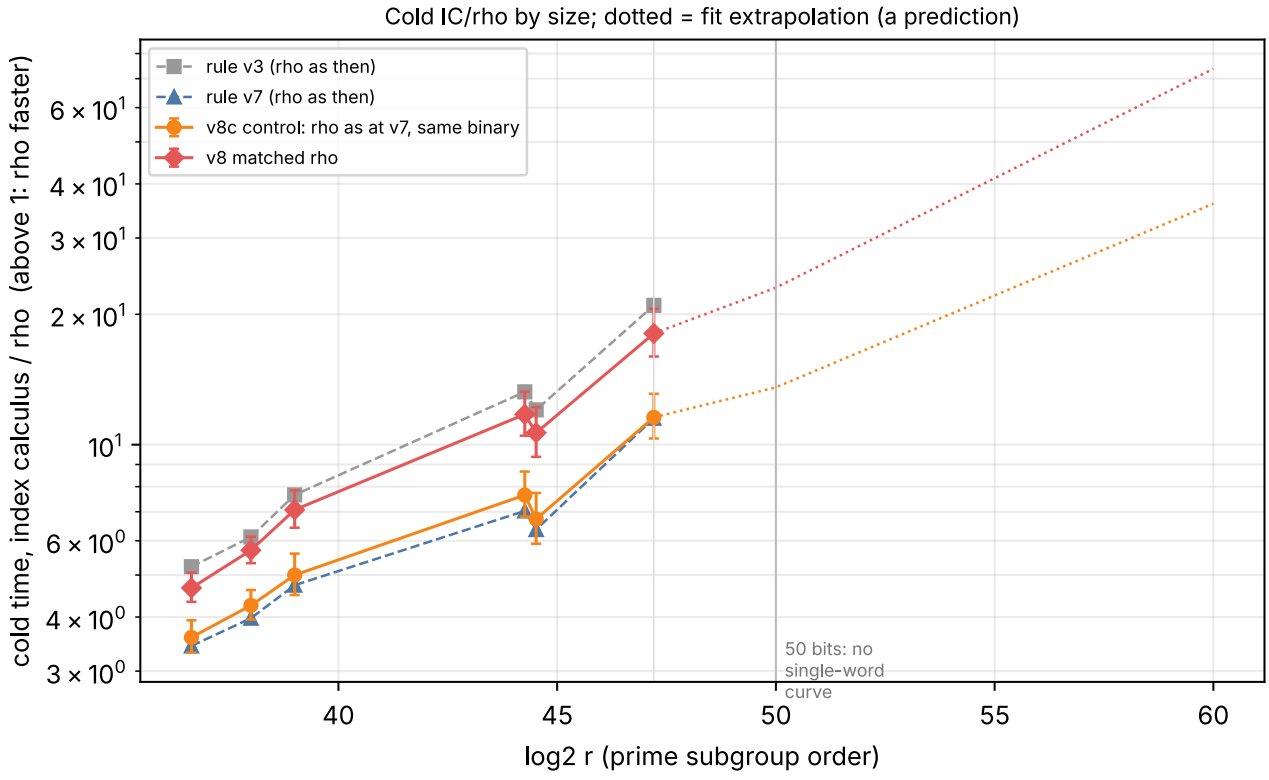
Checks

408 of 408 rows pass in both runs; every logarithm is §23's; on 64 of 64 paired targets at every size the two rho arms take the same steps and recover the same logarithms. The pin failed once before any timing (a counter, `row_ops`, added to main after §23): Amendment 1. The matched rho is admissible against the strong walk at all six sizes (0.57–0.82 of its time a step); the control is not at four of six (1.05, 1.20, 1.04, 1.03).

Cold IC/rho, with 95% bootstrap intervals over targets

log2 r	n	v8 matched	v8c control	v7	v3	matched / control
36.64	47	4.67 [4.34, 5.07]	3.59	3.43	5.23	1.30 [1.29, 1.31]
38.0	57	5.70 [5.33, 6.13]	4.25	3.97	6.12	1.34 [1.31, 1.37]
39.0	41	7.07 [6.43, 7.84]	5.00	4.74	7.66	1.41 [1.38, 1.45]
44.26	53	11.75 [10.49, 13.25]	7.65	7.03	13.24	1.54 [1.49, 1.58]
44.52	59	10.65 [9.38, 12.21]	6.74	6.37	12.03	1.58 [1.53, 1.64]
47.21	61	18.07 [15.98, 20.58]	11.56	11.51	20.97	1.56 [1.51, 1.62]

All six values lie inside the pre-declared prediction ranges. The index calculus's own online time, v8 over v8c, is 0.976–1.033: the measurement's drift is about 3%.



rule v8, 64 targets a size, 95% bootstrap over targets; Emerald Rapids container; source: matched.json, history.json

Slope of ln(cold ratio) on ln r

arm	slope	95% t (six sizes)	bootstrap over targets
v3	0.1756	0.137, 0.214	
v7	0.1440	0.092, 0.196	
v8c control	0.1409	0.096, 0.186	0.125, 0.157

v8 matched	0.1679	0.128, 0.208	0.152, 0.184

Kill condition 2 (slope not below v3's 0.1756) is not triggered: 0.1679 is below, "within the interval" (difference from v3 −0.008). Matching rho moved the slope by +0.027; the fall from v3 to v7 was mostly an unmatched-rho artefact. n and log2 r are not independent in these six curves. Condition 1 (gap at 50 bits or more) was not testable: no Koblitz curve with n ≤ 63 reaches 50 bits (largest 47.21) and `ic price` has no two-word path; the fit extrapolates 23.0× [18.3, 29.0] at 50 bits for the matched arm (a prediction).

